

Notes: Nurisso (JFE, 2026)

Learning by Lending Securities

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1 Big picture

- **Question.** What happens when securities lenders learn information from short sellers during the lending process, and how does that change lending fees, recall behavior, short-sale constraints, and price efficiency?
- **Setting.** A securities lender and an informed short seller interact in an over-the-counter lending market before trading against a Kyle-style market maker. The lender can set lending fees, infer information from borrowing demand, and in some cases recall shares before the short seller can sell.
- **Core friction.** Lending is not just a revenue source. A short seller's decision to borrow is itself an informative signal. This creates a new interaction between lending and trading: a lender may want to lower fees to learn more, but once he learns bad news he may want to front-run the short seller or strategically recall the loan.
- **Main result 1.** Lower lending fees improve the information content of the lending market signal because more negatively informed short sellers choose to borrow. In that sense, fees control the precision of the lender's signal.
- **Main result 2.** If the lender can recall shares, low fees are not sustainable. The lender is tempted to recall shares after learning that the stock is overvalued so that he can sell at a better price. Anticipating this, short sellers will not borrow unless the fee is high enough to make no-recall behavior incentive compatible.
- **Main result 3.** Passive lenders (such as index funds) charge *higher* fees than active lenders, but they can still attract *more* shorting demand because they do not trade on the information revealed in lending and do not strategically recall shares for trading purposes.
- **Main result 4.** Passive lending can improve price efficiency by relaxing short-sale constraints, especially when recall risk makes active lenders raise fees and discourage shorting.
- **Competition result.** Competition between passive lenders drives fees toward zero in the textbook Bertrand way. Competition between active lenders does much less because active lenders are already choosing the lowest feasible fees consistent with their incentives. Competition between a passive and an active lender can still leave the passive lender winning business at a higher fee because the passive lender imposes lower information-leakage costs on the short seller.
- **Why the paper matters.** The paper gives a tractable equilibrium framework for studying information leakage in securities lending. It connects active versus passive ownership, short-sale constraints, recall rules, and price efficiency in one model.

2 Data and measurement (key objects)

The paper is primarily theoretical. So in place of an empirical dataset, the relevant “data” are the objects and signals inside the model.

2.1 Core traded object and agents

- There is one stock with terminal payoff

$$v \in \{0, 1\}, \quad \Pr(v = 1) = \rho \in (0, 1).$$

- There are four agents:
 - an informed speculator / short seller S ,
 - a securities lender L ,
 - noise traders,
 - a competitive market maker.
- All agents are risk neutral.

2.2 Information structure and private objects

- The short seller observes the realization of v .
- She also draws an opportunity cost of trading c , interpreted as the value of her next-best use of capital.
- The distribution of c depends on the state. The paper imposes two simplifying assumptions:
 - when $v = 1$, $\bar{C}_1$ is low enough that the short seller always buys;
 - when $v = 0$, $\bar{C}_0$ is chosen so that if she already owned a share, she would always sell. In the paper this benchmark is pinned down by

$$\bar{C} = \frac{3\rho}{5}.$$

- The lender does *not* initially observe v , but he learns from whether the short seller borrows a share. The lending-market signal is

$$b \in \{1, 0\},$$

where $b = 1$ means the short seller borrows and $b = 0$ means she does not.

2.3 Trading quantities and order flow

- The lender's asset demand is

$$D_L \in \{-1, 0, 1\},$$

where -1 is a sale, 0 is no trade, and 1 is a purchase.

- The short seller's demand is

$$D_S \in \{-1, 0, 1\},$$

where $D_S = -1$ is a short sale. She must borrow in order to choose that action.

- Aggregate noise demand is

$$D_{NT} \in \{-2, -1, 0, 1, 2\},$$

with each outcome equally likely.

- Aggregate order flow is

$$z \equiv D_S + D_L + D_{NT}.$$

2.4 Timing and institutional mapping

The paper’s timing is crucial:

- First, the stock payoff v is drawn.
- Second, the short seller learns (v, c) .
- Third, in the lending stage, the lender offers a fee $f \geq 0$ and the short seller decides whether to borrow.
- Fourth, the lender observes whether borrowing occurred and updates his belief.
- Fifth, if recall is allowed, the lender can recall the share before trading. If he recalls, the fee is returned and the short seller cannot sell.
- Sixth, all traders submit orders to the market maker.
- Finally, v is realized publicly and short sales are settled.

The institutional motivation for recall is U.S. securities-lending practice, especially the rule that mutual funds must be able to recall lent-out shares at will. This makes recall risk economically relevant rather than a purely theoretical possibility.

2.5 No reduced-form dataset or estimation target

- The paper does not estimate the model structurally on micro data.
- Instead, the model is built to interpret and organize recent empirical evidence showing that active lenders trade on lending information, that passive ownership is associated with higher lending fees, and that passive ownership can relax short-sale constraints.

3 Models and empirical specifications (all equations + interpretation)

3.1 Market-maker pricing rule

The market maker sets price equal to expected value conditional on total order flow:

$$P(z) \equiv \mathbb{E}[v \mid z]. \tag{1}$$

- **Interpretation.** This is the Kyle-style pricing equation. Prices react only to net order flow, not to the lending market directly.
- **Why it matters.** Because the market maker does not observe borrowing demand, the lender can potentially profit from what he learns in the lending stage.

3.2 Passive lender benchmark

When the lender is passive, he never trades, so $D_L = 0$ regardless of the signal b .

The short seller's cutoff rule under a given fee f is

$$c(f) = \mathbb{E}[P(z) \mid D_S + D_L = -1] - f. \quad (3)$$

She shorts when $v = 0$ if and only if $c < c(f)$.

The passive lender chooses the fee to maximize expected lending revenue:

$$\max_f \Pr(D_S = -1)f = \Pr(v = 0) \left(\frac{\mathbb{E}[P(z) \mid D_S + D_L = -1] - f}{\bar{C}} \right) f. \quad (4)$$

The optimal fee is

$$f^* = \frac{\mathbb{E}[P(z) \mid D_S + D_L = -1]}{2} = \frac{3\rho}{10}. \quad (5)$$

- **Interpretation of (3).** The short seller compares expected selling profits net of the fee with her opportunity cost. This is just a threshold-entry condition.
- **Interpretation of (4).** The passive lender acts like a monopolist in a lending market with downward-sloping demand.
- **Interpretation of (5).** The passive lender extracts half of the short seller's expected selling profits. Fees are higher when short sales are rarer but more profitable, that is, when ρ is high and bad states are less likely.

3.3 Active lender without recall risk

Once the lender is allowed to trade on the lending signal, the short seller's cutoff becomes

$$c(f) = \mathbb{E}[P(z) \mid D_S + D_L = -2] - f. \quad (6)$$

The lender's posterior valuation after a loan agreement is

$$\mathbb{E}[v \mid b = 1] = 0, \quad (7)$$

while after no loan agreement it is

$$\mathbb{E}[v \mid b = 0] = \frac{\rho}{\rho + (1 - \rho)(1 - G(c(f)))}. \quad (8)$$

The lender now chooses the fee to maximize expected lending plus trading profits:

$$\max_f \Pr(D_S = -1) \left(\mathbb{E}[P(z) \mid D_S + D_L = -2] + f \right) + (1 - \Pr(D_S = -1)) D_L(b = 0) \mathbb{E}[v - P(z) \mid b = 0, D_L(b = 0)]. \quad (9)$$

For any candidate trading action after no loan agreement, the fee mapping is

$$f^* = \begin{cases} \max \left\{ -\frac{\mathbb{E}[P(z) \mid D_S + D_L = 1]}{2}, 0 \right\} = 0, & \text{if } D_L(b = 0) = 1, \\ 0, & \text{if } D_L(b = 0) = 0, \\ \max \left\{ \frac{\mathbb{E}[P(z) \mid D_S + D_L = -1]}{2}, 0 \right\} = \frac{\mathbb{E}[P(z) \mid D_S + D_L = -1]}{2}, & \text{if } D_L(b = 0) = -1. \end{cases} \quad (10)$$

The paper's main equilibrium result without recall risk is:

- the lender always sells after a loan agreement;
- after no loan agreement, both *buying* and *not trading* can be equilibrium strategies;
- selling after no loan agreement is never an equilibrium strategy;
- equilibrium fees are zero.
- **Economic intuition.** The active lender is willing to give up lending revenue in order to improve the informativeness of the borrowing signal. If he expects to buy after no loan agreement, he especially wants low fees because low fees make “no loan” a more informative sign that the stock is good. In that sense, the lender subsidizes short selling in order to learn.
- **Multiplicity.** The buying-versus-no-trade multiplicity comes from how prices react to the lender’s own expected trading behavior. If the market maker expects the lender to buy, prices move in a way that can validate buying; if he expects no trade, that can validate no trade.

3.4 Why standard active-lender logic breaks with recall risk

If the lender can recall shares after observing borrowing demand but before orders hit the market, he may want to recall the loan and prevent the short seller from selling.

The no-recall incentive-compatibility condition is

$$\mathbb{E}[P(z) \mid D_S + D_L = -2] + f \geq \mathbb{E}[P(z) \mid D_S + D_L = -1],$$

which implies

$$f \geq \Delta \mathbb{E}[P(z)] \equiv \mathbb{E}[P(z) \mid D_S + D_L = -1] - \mathbb{E}[P(z) \mid D_S + D_L = -2]. \quad (11)$$

- **Interpretation.** If the lender does not recall, he sells at the price associated with both parties selling and keeps the fee. If he recalls, he loses the fee but sells at a higher price because the short seller no longer trades. The fee must therefore compensate him for giving up the strategic-recall gain.
- **Key insight.** Recall risk creates a *positive lower bound* on feasible fees. Unlike the no-recall benchmark, the lender cannot push fees all the way to zero if he wants short sellers to keep borrowing.

3.5 Active lender with recall risk

With recall risk, the equilibrium changes sharply. The unique equilibrium has:

- the lender selling after a loan agreement,
- the lender *not trading* after no loan agreement,
- equilibrium fees equal to the minimum commitment fee,

$$f^* = \Delta \mathbb{E}[P(z)].$$

A direct corollary is that the active lender’s expected profits fall when recall risk is introduced.

- **Interpretation.** The lender is forced away from the fee that would maximize profits absent recall risk. The ability to recall shares hurts him in equilibrium because the short seller anticipates it and the lender must distort his fee upward to restore credibility.

3.6 Comparing passive and active equilibria

The paper then compares three cases: passive lending, active lending without recall risk, and active lending with recall risk.

The paper shows:

- passive lenders charge *higher* fees than active lenders;
- nevertheless, with recall risk, passive lenders attract *more* shorting demand than active lenders;
- without recall risk, passive lenders attract more shorting demand only when the active lender buys after no loan agreement.
- **Economic intuition.** An active lender imposes two costs on the short seller: the fee and the information-leakage / front-running cost. A passive lender imposes only the fee. Even though passive fees are higher, the total cost of borrowing can be lower with a passive lender.

3.7 Price efficiency

The paper defines price efficiency as

$$\text{Eff} \equiv 1 - \mathbb{E}[|v - \mathbb{E}[P(z) | v]|]. \quad (12)$$

- **Interpretation.** This is one minus the expected pricing error conditional on the true state. Higher values mean prices are closer, on average, to fundamental value.
- **Main result.** Without recall risk, price efficiency is always higher under the active lender because he extracts more information. With recall risk, price efficiency can be higher under the passive lender when ρ is sufficiently low, that is, when bad states are sufficiently likely and the relaxation of short-sale constraints matters more.

3.8 Extension 1: lender reputation in a repeated game

In an infinitely repeated version of the game, the lender may sustain no-recall behavior through reputation. The reduced-form condition is

$$f + \mathbb{E}[P(z) | D_S + D_L = -2] + \frac{\delta}{1 - \delta} \pi > \mathbb{E}[P(z) | D_S + D_L = -1] + \frac{\delta}{1 - \delta} \bar{\pi},$$

so that

$$f > \Delta \mathbb{E}[P(z)] - \frac{\delta}{1 - \delta} (\pi - \bar{\pi}). \quad (13)$$

- Here δ is the lender's discount factor, π is expected continuation profit when he has never recalled shares, and $\bar{\pi}$ is expected continuation profit once he has lost that reputation.

- **Interpretation.** Reputation lowers the fee necessary to sustain commitment. The more the lender values future business, the less he needs to rely on high current fees as a commitment device.

3.9 Extension 2: lender selling recalled shares

If the lender can sell recalled shares before physically regaining them, the commitment problem becomes more severe.

- The lender can now recall and dump the short seller's inventory into the market himself.
- In the paper's illustrative example, the fee needed to prevent this behavior becomes so high that short selling fully unravels in equilibrium.
- **Economic intuition.** Once recall allows the lender to appropriate essentially all of the short seller's trading profits, the short seller will not participate unless fees leave her enough surplus. In the extreme, that is impossible.

3.10 Extension 3: informed active lender

The lender may also receive a private signal $y \in \{0, 1\}$ about the payoff, with

$$\Pr(y = v \mid v) = q \in [0.5, 1].$$

The paper's key result is that the lender's private signal does not directly change the fee formula as a function of prices, but it changes which equilibrium fee is realized because a negatively informed lender is more inclined to sell and therefore to set a high fee. In particular, the negatively informed lender always sets a fee at least as high as the positively informed lender, and strictly higher whenever he still attracts short sellers.

- **Interpretation.** High fees can reflect not only strong shorting demand but also the lender's own bad information.

3.11 Extension 4: competition between lenders

The paper studies Bertrand fee competition in three cases.

- **Active vs. active.** Competition has essentially no effect on fees. Without recall risk, active lenders already set fees at zero. With recall risk, they already set the lowest feasible fee consistent with no recall.
- **Active vs. passive.** The passive lender can still win even at a higher fee because the short seller cares about the total cost of borrowing, not just the explicit fee. The passive lender imposes no trading-related information leakage.
- **Passive vs. passive.** Standard Bertrand logic applies and fees are driven to zero.

4 Identification and instruments (what makes the estimates credible)

This paper is a theoretical equilibrium paper, so there are no external instruments in the empirical sense. The relevant question is instead: what assumptions isolate each mechanism, and what exactly is doing the work?

4.1 What fails in the standard view of securities lending

- Standard securities-lending models usually treat the lender as passive and focus on fee revenue only.
- That benchmark misses the fact that borrowing demand itself is informative and that lenders may trade on this information.
- Once the lender is active, fee setting can no longer be understood as pure monopoly pricing.

4.2 What identifies the learning channel

- The market maker observes aggregate order flow but not the lending market directly.
- The short seller is perfectly informed about bad states, so a borrowing decision is informative.
- The lender observes borrowing before trading, so he can condition his portfolio choice on that signal.
- Together, these assumptions make the lending signal valuable and create the incentive to use fees to manipulate its informativeness.

4.3 What identifies the recall-risk channel

- Recall must occur after the lender sees borrowing demand but before the short seller can complete the short sale.
- The short seller must anticipate recall and refuse to borrow from lenders who cannot credibly commit not to recall.
- These assumptions are what generate the positive lower bound on fees in equation (11).

4.4 Why passive versus active lenders are a clean comparison

- Passive and active lenders differ in one key dimension: the ability to trade on information learned in the lending market.
- Holding everything else fixed, that isolates the information-leakage cost imposed on short sellers.
- This is why the comparison is useful for thinking about index funds versus active funds in securities lending.

4.5 Why the competition results are informative

- The competition section is not about generic Bertrand logic; it is about whether fees are above the lender's privately optimal floor.
- Active lenders are often already at their own lower bound, so competition does little.
- Passive lenders, by contrast, can price above marginal cost purely because of market power, so competition bites much more strongly.

4.6 Relation to the literature

- The paper relates to securities-lending models with lender market power, but differs by allowing lenders to trade on the information revealed by lending.
- It also connects to the passive-investing literature by showing that passive funds can improve price efficiency through the lending market even if they do not trade actively themselves.
- More broadly, it contributes to over-the-counter market microstructure by treating lending as an information acquisition stage that precedes trading.

5 Tables and figures

The main text is figure-driven rather than table-driven. The figures summarize the equilibrium forces very clearly.

5.1 Figure 1: Equilibrium prices

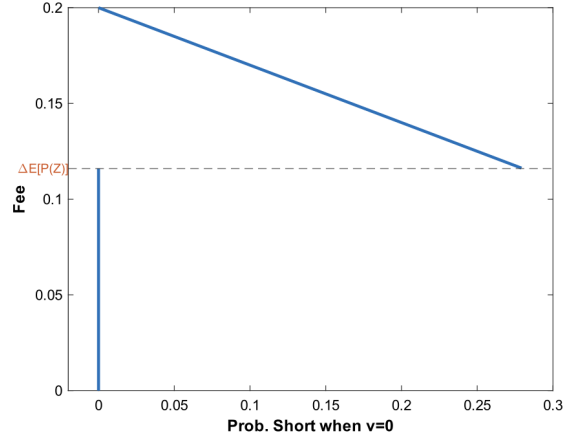
Read:

- The figure shows how the mapping from order flow to price changes depending on whether the active lender buys or does not trade after no loan agreement.
- When the lender buys after no loan agreement, some order flows that previously signaled bad news now also occur in good states, so those prices rise.
- This is the source of equilibrium multiplicity: the lender's own expected trade changes the informativeness of order flow.
- Economically, the figure shows that the lender's trading behavior affects how much value he can extract from the lending signal.

5.2 Figure 2: Lending-market demand curve

Read:

- The figure plots the probability of a short sale when $v = 0$ against the fee for an active lender with recall risk.



- Above the no-recall lower bound, lower fees attract more shorting demand in the usual downward-sloping way.
- Below the lower bound, demand collapses to zero because the lender can no longer commit not to recall shares.
- This is the clean visual statement of the paper's commitment logic.

5.3 Figure 3: Equilibrium fees

Read:

- The passive lender always charges the highest fee.
- The active lender without recall risk drives fees down to zero to maximize learning.
- Once recall risk is introduced, the active lender must raise fees to $\Delta\mathbb{E}[P(z)]$, but these fees still remain below passive-lender fees.
- The figure shows that the fee ranking comes from differences in objectives: pure revenue maximization for passive lenders versus information acquisition plus commitment for active lenders.

5.4 Figure 4: Equilibrium reduction in short selling

Read:

- The figure plots the equilibrium reduction in shorting, $1 - G(c^*)$, which can be read as a measure of short-sale constraints.
- With recall risk, active lenders deter more shorting than passive lenders because they must charge higher commitment fees and also leak information through trading.

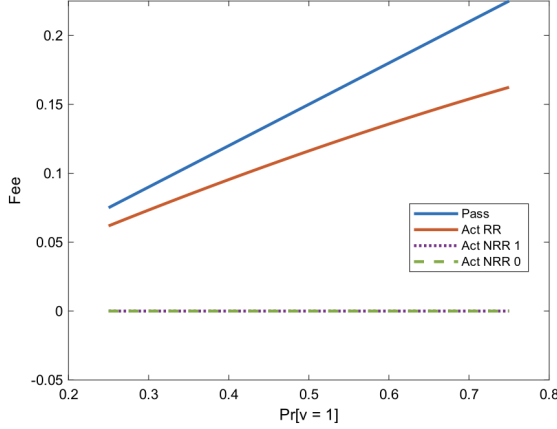


Fig. 3. Equilibrium Fees.

Fig. 3 plots the equilibrium fee f^* against the probability that $v = 1$ (i.e., ρ). The blue line represents the passive lender ("Pass"), the solid orange line represents the active lender with recall risk ("Act RR"), the dotted purple line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 1$ ("Act NRR 1"), and the dashed green line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 0$ ("Act NRR 0"). I also set $\bar{C} = \frac{3\rho}{5}$.

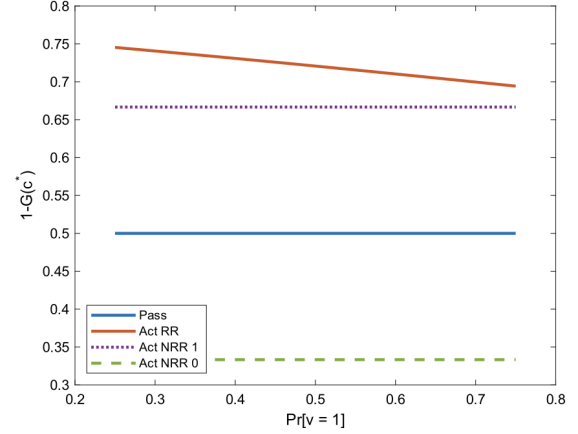


Fig. 4. Equilibrium Percentage Reduction in Short Selling.

Fig. 4 plots the equilibrium reduction in short selling $1 - G(c^*)$ against the probability that $v = 1$ (i.e., ρ). The blue line represents the passive lender ("Pass"), the solid orange line represents the active lender with recall risk ("Act RR"), the dotted purple line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 1$ ("Act NRR 1"), and the dashed green line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 0$ ("Act NRR 0"). I also set $\bar{C} = \frac{3\rho}{5}$.

- Without recall risk, the comparison depends on what the active lender does after no loan agreement.
- The figure clarifies that higher passive-lender fees do not imply tighter short-sale constraints once information leakage is accounted for.

5.5 Figure 5: Passive versus active price efficiency

Read:

- The figure plots the difference in price efficiency between passive and active lenders.
- Without recall risk, active lenders always improve price efficiency because they learn and trade on the signal.
- With recall risk, passive lenders can dominate when bad states are sufficiently likely, because they relax short-sale constraints enough to offset the loss of active informed trading.
- This figure delivers the paper's most novel asset-pricing implication: passive lending can improve prices even though passive lenders themselves do not trade on information.

6 Short summary

- **Conclusion.** Securities lending is not a side business detached from trading. It is an information acquisition technology, and that fact changes equilibrium fees and short-sale constraints.
- **Main mechanism.** Active lenders want low fees to learn, but recall risk forces them to keep fees high enough to commit not to exploit what they learn.

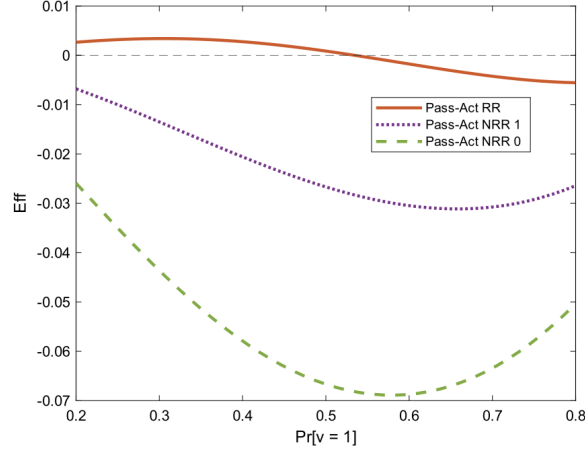


Fig. 5. Equilibrium Price Efficiency.

Fig. 5 plots the difference in price efficiency when the lender is passive vs. active. The solid orange line represents the difference in Eff between the passive lender and the active lender with recall risk (“Pass-Act RR”), the dotted purple line represents the difference in Eff between the passive lender and the active lender with no recall risk and active asset demand $D_L^*(b=0)=1$ (“Pass-Act NRR 1”), and the dashed green line represents the difference in Eff between the passive lender and the active lender with no recall risk and active asset demand $D_L^*(b=0)=0$ (“Pass-Act NRR 0”). I also set $\bar{C} = \frac{3p}{5}$.

- **Main implication.** Passive lenders can attract more shorting and may improve price efficiency, despite charging higher fees, because they do not impose trading-related information leakage on short sellers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that optimal securities-lending fees cannot be understood without considering the lender’s trading incentives.
- It shows that strategic recall creates a new commitment problem, yielding a positive lower bound on active-lender fees.
- It also shows that passive funds’ inability to trade on lending information can be a feature rather than a bug from the perspective of short-sale constraints and, in some cases, price efficiency.

7.2 Economic intuition

The paper’s intuition is simple but powerful. Borrowing demand is bad news. An active lender would like to extract as much information from that bad-news signal as possible, which pushes him toward low fees. But once the bad news is revealed, he would also like to use it before the short seller can do so. Recall rights let him do exactly that unless the fee is high enough to make restraint worthwhile. Passive lenders avoid this problem because they cannot exploit the signal through trading. As a result, they may become the preferred counterparties for short sellers even while charging more.

7.3 Takeaway

This paper is best read as a theory of how information leakage in the securities-lending market shapes asset prices. Its central message is that the rise of passive investing changes short-sale constraints not only through greater lendable supply, but also through who can credibly commit not to exploit lending information. That insight helps organize a growing empirical literature on active lending, passive ownership, borrowing fees, and price efficiency.